LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	18
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Level flight performances, power available and required for jet and propeller aircrafts. Best climb speeds.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Level flight performances, power available and required for jet and propeller aircrafts. Best climb speeds.
5. Specific Objectives	11.1.3.1 Trainee will be able to describe flight performance of aircraft.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of theoretical graphs; Instructor supervision required for safety in hypothetical scenarios.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the definition of level flight?
(b) How does power available compare to power required in steady level flight?
(c) Name one factor affecting best climb speed in propeller aircraft.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Power Available vs Power Required Curves • Excess Power Definition and Significance • Jet Aircraft Performance Characteristics • Propeller Aircraft Performance Characteristics • Relationship between Excess Power and Climb Rate	L & Ds	PPT, IB	15 min
(b)	• Best Angle of Climb Speed (Vx) • Best Rate of Climb Speed (Vy) • Differences between Jet and Propeller Vx/Vy • Impact of Weight on Climb Performance • Impact of Density Altitude on Climb Performance	L & Ds	PPT, PDF	20 min
(c)	• Calculating Time to Climb • Calculating Distance to Climb • Practical Application of Climb Speeds • Safety Margins in Climb Performance • Transition from Level to Climb Flight	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between excess power and rate of climb.
(b) Differentiate between Vx and Vy for a propeller aircraft.
(c) How does increasing aircraft weight affect best climb speeds?

12. Home Assignment and Follow Up	Prepare a brief note comparing the power required curves of a jet and a propeller aircraft, highlighting how these differences influence their respective best climb speeds, to be discussed in LP 19.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Forces in climb, Excess power, max angle, maximum rate, Economical climb, time to climb.	Refer to the master syllabus reference list	LP 19

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min